

ORF309/MAT380

**Barnoulli Processes III: Law of
Large Numbers**

(pp.47-54 (optional pp.54-56))

Ioannis Akrotirianakis

Agenda for today

- ***Law of Large Numbers***
 - *Probabilities as frequencies (or fractions...)*
- Expectation and Conditional Probabilities revisited
- **Variance**
 - Measure of randomness

Intuitive notion of probability

- **Our intuition says:**
 - If we perform a random experiment many times, the **probability** of an event A is the **fraction** of the experiments in which A occurs.

$$P(A) = \frac{\text{\# of experiments where } A \text{ occurs}}{\text{total \# of experiments}}$$

- But this definition is **not so clear**:
 - How many experiments should we perform?
 - Different number of experiments will always give you different $P(A)$

Superexperiment

- Let's define a **Superexperiment** (SE)



*(which is also Probability Model,
with its own assumptions)*

- We have an Original Experiment (OE) and an event A with $P(A) = p$
- We repeat the OE many times within the SE
- In each repetition, event A occurs with probability p
- We also define the events: $A_i = \{\text{event } A \text{ occurred in repetition } i\}$
- $A_i \perp\!\!\!\perp A_j \quad \forall i, j = 1, 2, 3, \dots$ and $P(A_i) = P(A) = p, \quad \forall i = 1, 2, 3, \dots$

Superexperiment and Bernoulli Process

- The Superexperiment provide a **perfect environment** to define the **event**:
(mathematical model)
 - “# of times event A happened in the first n experiments”
- But this event is something very familiar → **Bernoulli Process**:
(sequence of Bernoulli RVs)

Superexperiment and Bernoulli Process

- The Superexperiment provide a **perfect environment** to define the **event**:
(mathematical model)
- “# of times event A happened in the first n experiments”
- But this event is something very familiar → **Bernoulli Process**:
(sequence of Bernoulli RVs)

$$S_n = 1_{A_1} + 1_{A_2} + \cdots + 1_{A_n} = \sum_{k=1}^n 1_{A_k} \sim \text{Binom}(n, p)$$

- **Note:** $P(1_{A_k}) = P(A_k) = P(A) = p$

Superexperiment and Bernoulli Process

- Let's now define this fraction

$$\frac{\text{\# of times event } A \text{ happened in the first } n \text{ experiments}}{\text{total \# of experiments}}$$

- This fraction can also be expressed as a Random Variable:

Superexperiment and Bernoulli Process

- Let's now define this fraction

$$\frac{\text{\# of times event } A \text{ happened in the first } n \text{ experiments}}{\text{total \# of experiments}}$$

- This fraction can also be expressed as a Random Variable:

$$\frac{S_n}{n} = \frac{1}{n} \sum_{k=1}^n 1_{A_k} \quad \text{with distribution: } P\left(\frac{S_n}{n} = x\right) = P(S_n = nx), \quad \forall x \in [0,1]$$

Superexperiment and Bernoulli Process

- Let's now define this fraction

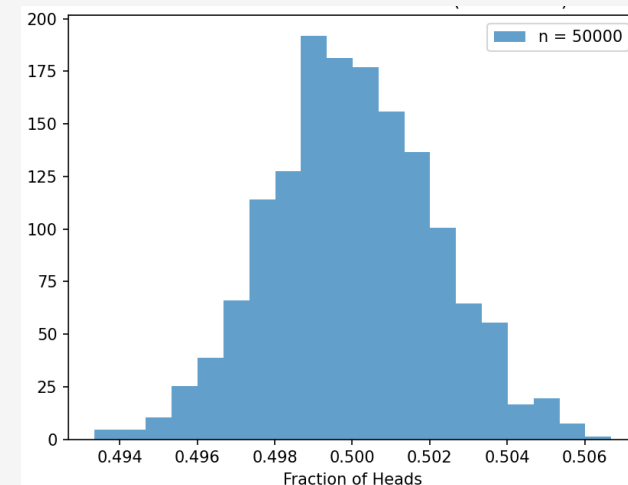
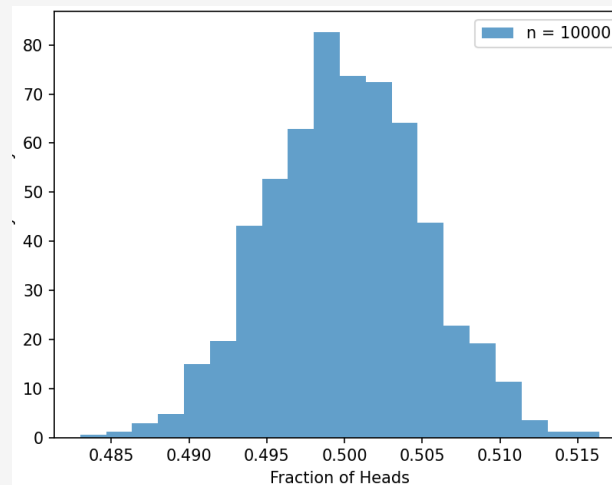
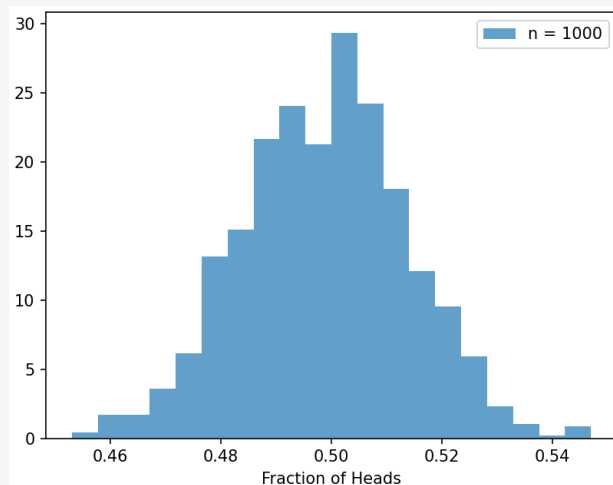
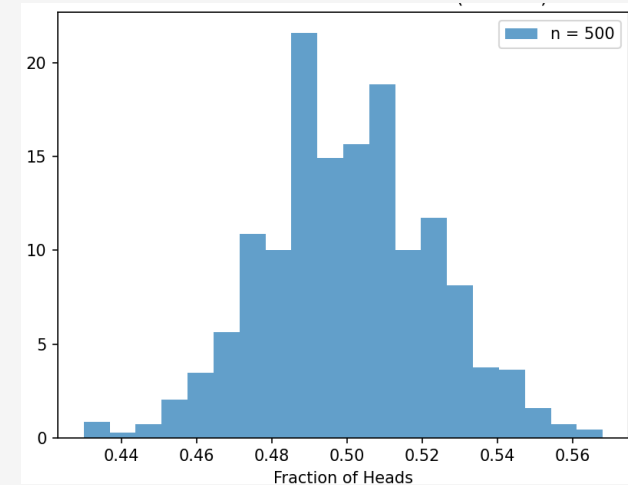
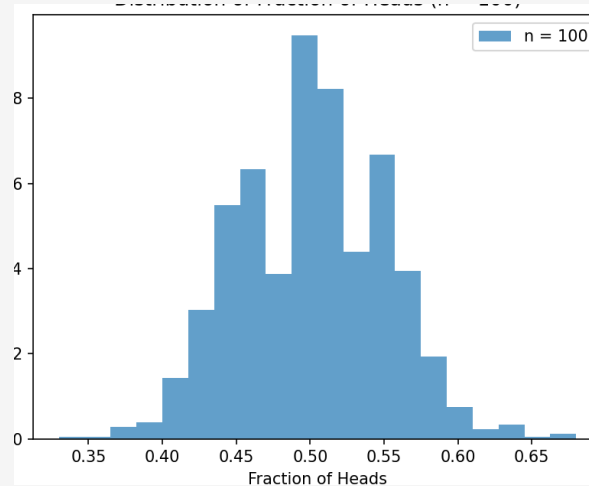
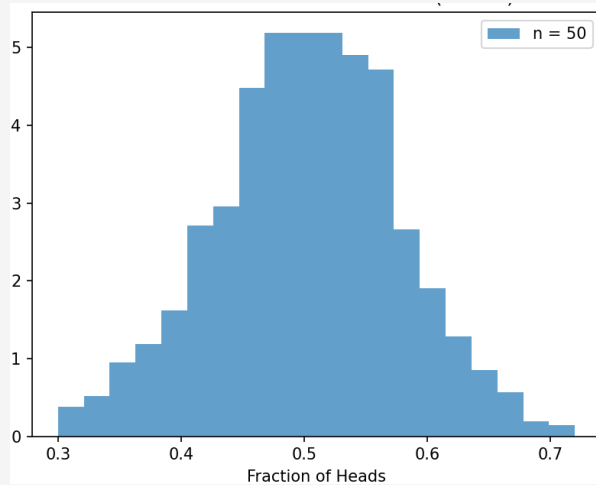
$$\frac{\text{\# of times event } A \text{ happened in the first } n \text{ experiments}}{\text{total \# of experiments}}$$

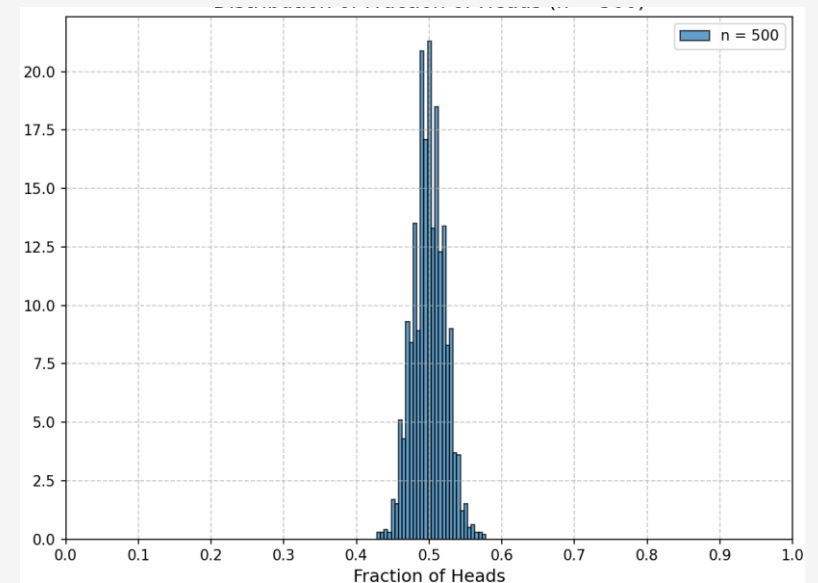
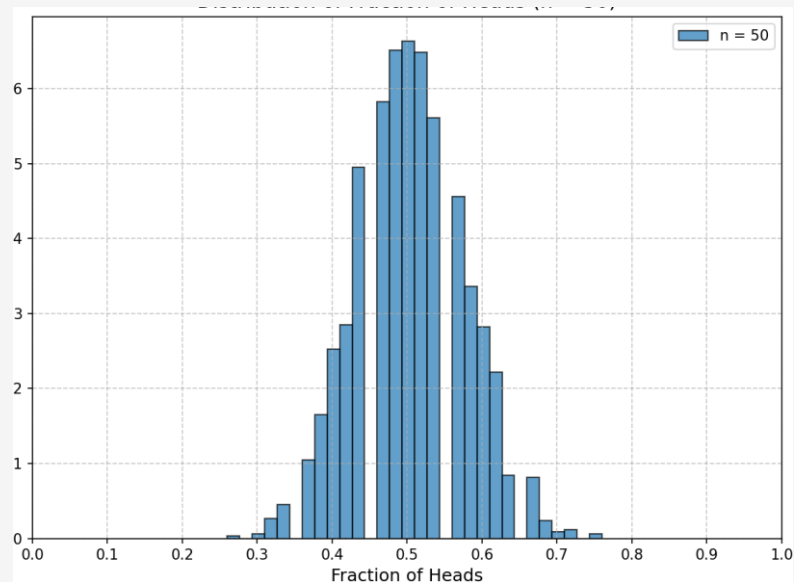
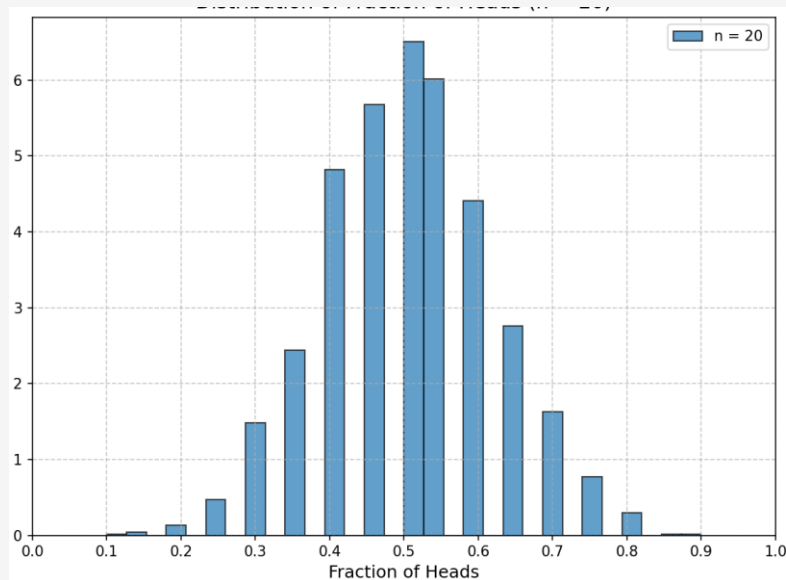
- This fraction can also be expressed as a Random Variable:

$$\frac{S_n}{n} = \frac{1}{n} \sum_{k=1}^n 1_{A_k} \quad \text{with distribution: } P\left(\frac{S_n}{n} = x\right) = P(S_n = nx), \quad \forall x \in [0,1]$$

Note! $\frac{S_n}{n}$ is a random variable that takes values in in the **interval** $[0,1]$

Let's run some Superexperiments





The histograms in this and the previous slide are for illustrative purposes only, demonstrating how the number of trials (n) affects the distribution of the random variable S_n/n . This represents the **fraction of heads obtained in a series of coin flip experiments**. In each experiment, a fair coin is flipped many times, and the fraction of heads (S_n/n) is recorded. The histograms show the distribution of these proportions across many experiments. As n increases, the values of S_n/n cluster more closely around $p=0.5$, which is the probability of getting heads on a single flip of a fair coin.

When the number of Superexperiments gets very large

- **Observation:** As we increase n the randomness of the fraction S_n/n gets closer and closer to the probability $P(A)$
- So, if we tend to take an infinite number of experiments (i.e, $n \rightarrow \infty$) then the distribution of the RV S_n/n will concentrate around $P(A) = p$

When the number of Superexperiments gets very large

- **Observation:** As we increase n the randomness of the fraction S_n/n gets closer and closer to the probability $P(A)$
- So, if we tend to take an infinite number of experiments (i.e, $n \rightarrow \infty$) then the distribution of the RV S_n/n will concentrate around $P(A) = p$
- In that case, $\lim_{n \rightarrow \infty} S_n/n$ is NOT random any more!

Expectation of S_n/n

- Since S_n/n is a random variable, let's compute **its expectation**

$$E\left[\frac{S_n}{n}\right] = E\left[\frac{1}{n} \sum_{k=1}^n 1_{A_k}\right] = \frac{1}{n} \sum_{k=1}^n E[1_{A_k}] = \frac{1}{n} \sum_{k=1}^n P(A_k) = \frac{1}{n} \sum_{k=1}^n P(A) = P(A) = p$$

- Hence the **expectation** of S_n/n is **constant** for any n , and is **always equal to p**
- **Remark**: Note that here we have **not** assumed that $n \rightarrow \infty$

Law of Large Numbers

Theorem of the Law of Large Numbers

- **Theorem:** With probability 1 we have:

$$\lim_{n \rightarrow \infty} \frac{S_n}{n} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n 1_{A_k} = P(A) = p$$

PROOF (after we introduce some more information $\rightarrow$ variance...)

What does it mean: “with probability 1:” ??

In finite number of experiments: “with probability 1” means certainty

In an infinite number of experiments: “with probability 1” means the event is guaranteed to happen in the long run

- **Remarks:**

- $\frac{1}{n} \sum_{k=1}^n 1_{A_k}$ is **RANDOM**, **for every finite** n (i.e., it is a random variable!)
- $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n 1_{A_k}$ is **NOT RANDOM!** (i.e., it is not a random variable!)
 - If we run the experiment ∞ times, then $P(A)$ is the **frequency** of A
 - If we run the experiment ∞ times, then $P(A)$ is **known**

Law of Large Numbers for Random Variables

- The idea of **Superexperiment** is also helpful here.
- Consider a random variable $X: \Omega \rightarrow D$
- Define the random variables X_i = "the outcome of X in the i -th experiment"
- The random variables X_i for $i = 1, 2, \dots$ have the same distribution as X
- **Theorem:** With probability 1 we have

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n X_k = E[X]$$

-
- **Proof:** We can write the DRV $X: \Omega \rightarrow D$ in terms of the **indicator** RV, as follows

$$X = \sum_{u \in D} u 1_{\{X=u\}} \quad \rightarrow \quad \begin{array}{l} \text{if } X = u \text{ for some } u \in D, \text{ then the} \\ \text{summation reduces to just one term:} \\ X = u \end{array}$$

- Then we can also write the X_k 's in terms of indicator RVs:

$$\frac{1}{n} \sum_{k=1}^n X_k = \frac{1}{n} \sum_{k=1}^n \sum_{u \in D} u 1_{\{X_k=u\}} = \frac{1}{n} \sum_{u \in D} \sum_{k=1}^n u 1_{\{X_k=u\}} = \frac{1}{n} \sum_{u \in D} u \sum_{k=1}^n 1_{\{X_k=u\}}$$

-
- Taking the limit as $n \rightarrow \infty$ yields:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n X_k = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{u \in D} u \sum_{k=1}^n 1_{\{X=u\}} = \sum_{u \in D} u \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n 1_{\{X=u\}} = ? ?$$

-
- Taking the limit as $n \rightarrow \infty$ yields:

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n X_k &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{u \in D} u \sum_{k=1}^n 1_{\{X=u\}} = \sum_{u \in D} u \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n 1_{\{X=u\}} = \\ &= \sum_{u \in D} u P\{X = u\} = E[X] \quad \blacksquare\end{aligned}$$

- So, **expectations** are indeed **averages**, only when $n \rightarrow \infty$.

LLN for conditional probability

- **Theorem:** With probability 1 we have

$$\lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n 1_{A_k \cap B_k}}{\sum_{k=1}^n 1_{B_k}} = \frac{P(A \cap B)}{P(B)} = P(A|B)$$

Proof: Exercise for you!

Remark: The reasoning and the proof is **very similar** as in the previous theorems. In this case we apply the LLN separately for the numerator and denominator

Variance (measure of randomness...)

Variance – a measure for randomness

- So, as $n \rightarrow \infty$ the random variable S_n/n becomes **increasingly less random...**
- But how can we measure randomness??
- **Definition:** Let $X: \Omega \rightarrow D$ be a random variable. Then the **variance** of X is defined as

$$\text{Var}(X) = E[(X - E[X])^2]$$

- **Remark:** a random variable becomes **less random** if it is **closer to its expectation** (equivalently, the **squared deviations of X from $E[X]$ become smaller**)

Properties of Variance

1. $\text{Var}(X) \geq 0$
2. $\text{Var}(X) = 0$ if and only if (iff) X is NOT random
3. $\text{Var}(X) = E[X^2] - (E[X])^2$
4. If $X \perp\!\!\!\perp Y$ then $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$
5. $\text{Var}(\alpha X) = \alpha^2 \text{Var}(X), \quad \forall \alpha \in R$

Proofs as Exercise!

Squared difference -vs- Absolute difference

- **Remark:** we defined the variance using the **squared error** between X and $E[X]$:

$$E[(X - E[X])^2]$$

- We could have used the **absolute error** between X and $E[X]$

$$E[|X - E[X]|]$$

- This is a perfect measure for measuring randomness. However, the 5 very convenient properties we defined earlier do not hold. When we discuss the [Central Limit Theorem](#), we will that the squared form is needed.

Proof of the LLN Theorem

- **Theorem:** With probability 1 we have:

$$\lim_{n \rightarrow \infty} \frac{S_n}{n} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n 1_{A_k} = P(A) = p$$

- **Proof:** we must show that S_n/n becomes **nonrandom** as $n \rightarrow \infty$.
In other words, **we need to show**

$$\lim_{n \rightarrow \infty} \text{Var} \left(\frac{S_n}{n} \right) = \lim_{n \rightarrow \infty} E \left[\left(\frac{S_n}{n} - p \right)^2 \right] = 0$$

-
- Let's compute the expectation (without the limit):

$$\begin{aligned} E \left[\left(\frac{S_n}{n} - p \right)^2 \right] &= E \left[\left(\frac{1}{n} \sum_{i=1}^n 1_{A_i} - E \left[\frac{1}{n} \sum_{i=1}^n 1_{A_i} \right] \right)^2 \right] = \text{Var} \left(\frac{1}{n} \sum_{i=1}^n 1_{A_i} \right) \quad \text{Property 5} \\ &= \frac{1}{n^2} \text{Var} \left(\sum_{i=1}^n 1_{A_i} \right) \quad \begin{array}{l} \text{Property 4} \\ (1_{A_i} \perp\!\!\!\perp 1_{A_j}) \end{array} = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(1_{A_i}) \end{aligned}$$

-
- Let's compute the expectation (without the limit):

$$E \left[\left(\frac{S_n}{n} - p \right)^2 \right] = E \left[\left(\frac{S_n}{n} - E \left[\frac{S_n}{n} \right] \right)^2 \right] = E \left[\left(\frac{1}{n} \sum_{i=1}^n 1_{A_i} - E \left[\frac{1}{n} \sum_{i=1}^n 1_{A_i} \right] \right)^2 \right] =$$

$$= \text{Var} \left(\frac{1}{n} \sum_{i=1}^n 1_{A_i} \right) \stackrel{\text{Property 5}}{=} \frac{1}{n^2} \text{Var} \left(\sum_{i=1}^n 1_{A_i} \right) \stackrel{\substack{\text{Property 4} \\ (1_{A_i} \perp\!\!\!\perp 1_{A_j})}}{=} \frac{1}{n^2} \sum_{i=1}^n \text{Var}(1_{A_i})$$

-
- Let's compute: $Var(1_{A_i})$

Property 3

$$Var(1_{A_i}) = E[(1_{A_i})^2] - (E[1_{A_i}])^2 = E[1_{A_i}] - (P(A_i))^2 = p - p^2 = p(1 - p)$$

- Substituting it into the equation of previous slide:

$$E\left[\left(\frac{S_n}{n} - p\right)^2\right] = \frac{1}{n^2} \sum_{i=1}^n p(1 - p) = \frac{1}{n^2} n p(1 - p) = \frac{1}{n} p(1 - p)$$

-
- Taking limits in both sides:

$$\lim_{n \rightarrow \infty} E \left[\left(\frac{S_n}{n} - p \right)^2 \right] = \lim_{n \rightarrow \infty} \frac{p(1-p)}{n} = 0$$

- Since $\left(\frac{S_n}{n} - p \right)^2 \geq 0$ we can safely conclude:

$$\lim_{n \rightarrow \infty} \left(\frac{S_n}{n} - p \right) = 0 \quad \Rightarrow \quad \lim_{n \rightarrow \infty} \frac{S_n}{n} = p$$



Summary of sampling methods

-
- When selecting a **sample of size** r from a **finite sample space**

$$\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$$

- of size n , **the number of possible outcomes** depends on whether the **order** of selection matters and whether the elements are **replaced** after our selection.
- In the lectures have covered most of them and we will also discuss them again and use them throughout the course.
- In the next slides we summarize the most important sampling methods.

Sampling without replacement and ordered (aka, permutations)

- It is used when we select r outcomes from Ω and the sequence is important, and each item can only be used once.
- The number of distinct ordered samples of size r is

$$N = n(n-1)(n-2) \dots (n-(r-1)) = \frac{n!}{(n-r)!}$$

- **Example:** we attend a horse race with 12 horses, and we want to know how many ways the horses can finish in 1st, 2nd, and 3rd place.

Sampling without replacement and unordered (aka, combinations)

- It is used when we select r outcomes from Ω and the sequence of the selected outcomes is not important, and each item can only be used once.
- The number of distinct ordered samples of size r is

$$N = \frac{n(n-1) \dots (n-(r-1))}{r(r-1) \dots 2} = \binom{n}{r}$$

- **Example:** when you play your favorite lottery you must choose 6 numbers from 49 different numbers.

Sampling with replacement, ordered

- It is used when you need to sample r outcomes and their order matter, and each time you select one you replace it (or return it) in the sample space.
- The number of distinct samples is

$$N = n^r$$

- **Example:** You want to create a pin of 7 digits (the order of numbers is important, and you can select the same digit many times)

Sampling with replacement unordered

- It used when you need to sample r outcomes and their order does not matter, and each time you select one you replace it (or return it) in the sample space.
- The number to choose r items from n possibilities is

$$N = \binom{n + r - 1}{r}$$

- **Example:** your favorite ice cream store has $n = 5$ unique flavors, and you want to buy $r = 3$ servings (scoops) to take home (*see next slide for more details*)

-
- To visualize this sampling technique and understand its formula, imagine you need to select r items (7 scoops) and you have n categories of items (20 flavors).
 - To separate n categories you need $n - 1$ bars.
 - Now consider the items you want to select as stars
 - To find the total number of combinations, you need to compute the number of ways you can arrange the total number of symbols in a row.
 - Or equivalently, the total number of ways you can choose r positions for the stars from $n + r - 1$ total available